SAT Math Mastery · Session 10

Circles: Equations, Arcs and Tangents

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	circle equations: completing the square, center and radius · writing a circle's equation from conditions · arc length and sector area · chords, diameters and inscribed angles · tangent lines and radii · lines meeting circles · inscribed polygons
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

In the xy-plane, the graph of $2x^2 - 8x + 2y^2 + 12y = 24$ is a circle. What is the radius of the circle?

(A) $\sqrt{50}$

(B) 5

(C) 6.5

(D) $\sqrt{40}$

2. HARD · DESMOS

A circular roundabout has its center at $(2, -5)$ in a coordinate model of the area, where distances are measured in hundreds of meters. Road k is tangent to the roundabout at the point $(9, -1)$. What is the slope of road k?

3. HARD · DESMOS

A circle has a radius of 12 meters. What is the area, in square meters, of a sector with a central angle of 150°? (Use $\pi \approx 3.14$.)

(A) 150.8

(B) 172.3

(C) 188.4

(D) 205.6

4. HARD · DESMOS

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 10y + 18 = 0$. What is the distance, in units, from the center of the circle to the origin?

(A) $\sqrt{34}$

(B) 34

(C) 4

(D) $2\sqrt{34}$

5. HARD · DESMOS

A circle in the xy-plane has center $(0, 3)$ and passes through the point $(5, 15)$. Which of the following is an equation of the circle?

(A) $(x-5)^2 + (y-3)^2 = 169$

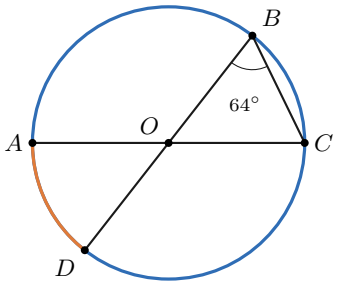
(B) $x^2 + (y-3)^2 = 13$

(C) $x^2 + (y-3)^2 = 144$

(D) $x^2 + (y-3)^2 = 169$

6. HARD · DESMOS

O is the center of a circle. Chords $\overline{AC}$ and $\overline{BD}$ are diameters that intersect at O (so A, O, C are collinear, and B, O, D are collinear). OB and OC are radii of the circle. If the measure of angle OBC is $64°$, what is the measure, in degrees, of arc AD?



7. HARD · DESMOS

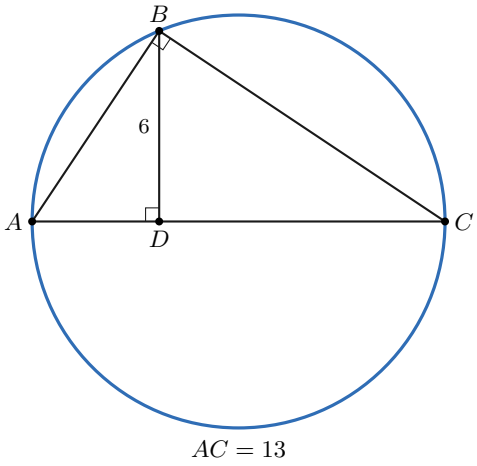
A sector of a circle with radius 9 has an area of 18π. The central angle of the sector measures $k\pi$ radians. What is the value of k?

8. HARD · DESMOS

The equation $x^2 + y^2 - 8x + 10y = 8$ defines a circle in the xy-plane. What is the radius of the circle?

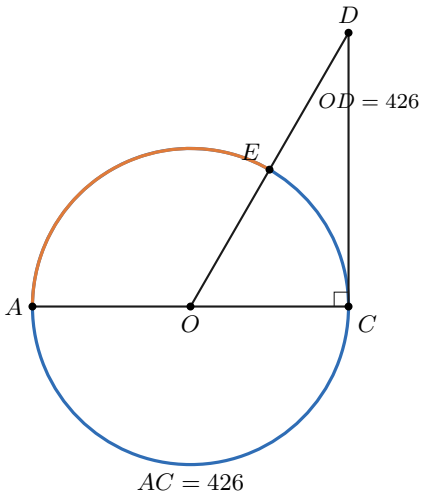
9. HARD · DESMOS

Points A, B, and C lie on a circle, and $\angle ABC$ is a right angle. Point D lies on segment AC such that $BD \perp AC$, with $BD = 6$. The diameter of the circle is 13 and $AB < BC$. What is the value of $\dfrac{CD}{AD}$?



10. HARD · DESMOS

In a circle with center O, segment AC is a diameter and $AC = 426$. Point D lies outside the circle with $OD = AC$, and $\overline{CD}$ is tangent to the circle at C. Segment OD intersects the circle at E. The length of minor arc AE is $b\pi$. What is the value of b?



11. HARD · DESMOS

Circle A in the xy-plane has equation $x^2 + y^2 - 8x + 6y - 8n = 0$, where n is a constant. Circle B has the same center as circle A, has twice the diameter of circle A, and passes through $(10, -3)$. What is the value of n?

12. HARD · DESMOS

In the xy-plane, a circle passes through the points $(-1, 2)$, $(7, 2)$ and $(6, 3)$. Which of the following is an equation of the circle?

(A) $(x-3)^2 + (y+1)^2 = 5$

(B) $(x-3)^2 + (y-1)^2 = 17$

(C) $(x-3)^2 + (y+1)^2 = 25$

(D) $(x-3)^2 + (y-2)^2 = 16$

13. HARD · DESMOS

In the xy-plane, the line $x + y = 4$ intersects the circle $x^2 + y^2 - 4x + 2y = 24$ at two points. What is the distance between these two points?

(A) 7

(B) $7\sqrt{2}$

(C) 14

(D) $\dfrac{7\sqrt{2}}{2}$

14. HARD · DESMOS

In the xy-plane, the line $y = k(x-1) - 2$, where k is a positive constant, is tangent to the parabola $y = x^2 + 2x + 4$. What is the value of k?

(A) -2

(B) 4

(C) 8

(D) 10

15. HARD · DESMOS

$$x^2 + y^2 - 6x + 8y = k$$

In the xy-plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x - 3y = -11$. What is the value of k?

16. HARD · DESMOS

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the xy-plane, the circle above is defined for each real number a. Which of the following points lies on the circle for every real value of a?

(A) $(a + 5,\ 2a + 3)$

(B) $(a + 5,\ 2a + 4)$

(C) $(2a + 5,\ 2a + 3)$

(D) $(a - 1,\ 2a + 2)$

17. HARD · DESMOS

$$(x-1)^2+(y+3)^2=40$$

$$x^2+y^2-4kx+2ky=9$$

In the xy-plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k?

(A) $1+2\sqrt{7}$ (B) $1+\sqrt{7}$

(C) $-1+\sqrt{7}$ (D) $2+\sqrt{7}$

18. HARD · DESMOS

In the xy-plane, a circle has equation $x^2+y^2+Dx+Ey+F=0$, where D, E, and F are constants. The circle passes through the points $(2,7)$ and $(10,7)$, has radius 5, and has its center below the line $y=7$. What is the value of $E+F$?

19. Hard · Desmos

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

In the xy-plane, the graph of the equation above is a circle. Line ℓ has slope $\frac{3}{4}$ and is tangent to the circle at a point with a positive y-coordinate. What is the y-intercept of ℓ?

(A) $-\frac{43}{4}$

(B) 1

(C) $\frac{7}{4}$

(D) $-\frac{1}{3}$

20. Hard · Desmos

In the xy-plane, the circle with equation $(x-7)^2 + (y+2)^2 = 9$ has two distinct tangent lines that pass through the point $(2, -2)$. What is the product of the slopes of these two lines?

21. Hard · Desmos

In the xy-plane, a circle lies in the first quadrant and is tangent to both the x-axis and the y-axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

22. HARD · DESMOS

In the xy-plane, the center of a circle lies on the y-axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

(A) $x^2 + (y+1)^2 = 40$

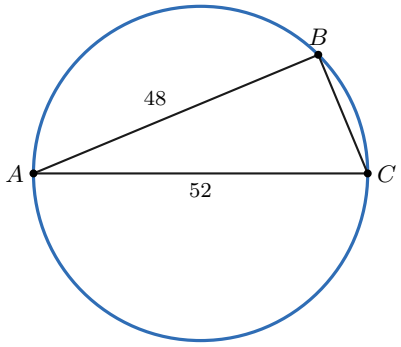
(B) $x^2 + (y-1)^2 = 40$

(C) $x^2 + (y+1)^2 = 20$

(D) $x^2 + (y+1)^2 = 44$

23. HARD · DESMOS

Triangle ABC is inscribed in a circle with circumference 52π. Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?



(A) $\frac{24}{13}$

(B) $\frac{5}{12}$

(C) $\frac{5}{13}$

(D) $\frac{10}{13}$

24. HARD · DESMOS

In the xy-plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2?

25. HARD · DESMOS

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q. Let C be the center of the circle. What is the area of triangle CPQ?

(A) $\frac{25}{2}$

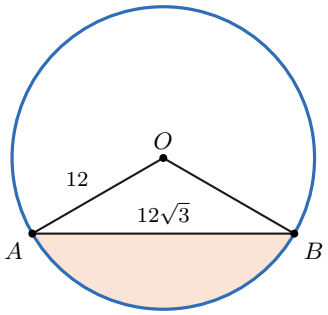
(B) 6

(C) 12

(D) 24

26. HARD · DESMOS

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB?



(A) $24\pi - 36\sqrt{3}$

(B) $48\pi - 36\sqrt{3}$

(C) 48π

(D) $96\pi + 36\sqrt{3}$

27. HARD · DESMOS

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

(A) $15 + 6\pi$

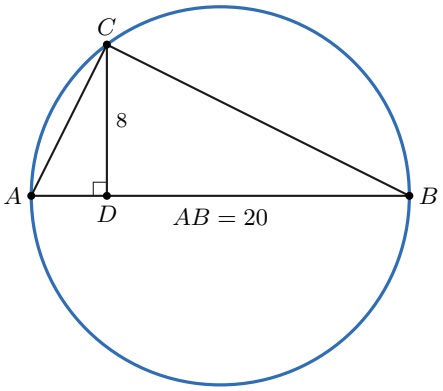
(B) $30 + 3\pi$

(C) $30 + 45\pi$

(D) $30 + 6\pi$

28. HARD · DESMOS

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the length of $\overline{AC}$?



(A) $4\sqrt{5}$

(B) $8\sqrt{5}$

(C) $4\sqrt{17}$

(D) 10

29. HARD · DESMOS

A circle has area 48π. A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

(A) $96 - 27\sqrt{3}$

(B) $96 - 36\sqrt{3}$

(C) $96 - 12\sqrt{3}$

(D) $192 - 36\sqrt{3}$

30. HARD · DESMOS

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k?

(A) 65

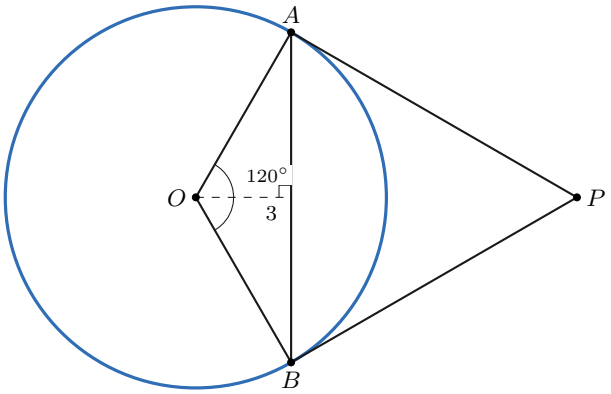
(B) 32

(C) 33

(D) -97

31. HARD · DESMOS

In a circle with center O, chord $\overline{AB}$ has $\angle AOB = 120°$ and lies at a perpendicular distance of 3 from O. The lines tangent to the circle at A and at B intersect at point P. What is the length of $\overline{PA}$?



(A) $6\sqrt{3}$

(B) 12

(C) $3\sqrt{3}$

(D) 6

32. HARD · DESMOS

In the xy-plane, a circle is tangent to both the x-axis and the y-axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2+y^2+Dx+Ey+F = 0$, where D, E and F are constants. What is the value of F?

33. HARD · DESMOS

$$x^2 + y^2 + 10x - 24y = 0$$

In the xy-plane, the graph of the equation above is a circle. What is the diameter of the circle?

34. HARD · DESMOS

In the xy-plane, a circle has its center on the line $y = 2x$ and passes through the points $(4, 14)$ and $(-1, -1)$. What is the square of the radius of the circle?

35. HARD · DESMOS

In the xy-plane, the circle $(x-2)^2+(y+1)^2=25$ passes through the point $(6,2)$. The line tangent to the circle at $(6,2)$ crosses the x-axis at the point $(a,0)$. What is the value of a?

(A) $\frac{26}{3}$

(B) $\frac{3}{2}$

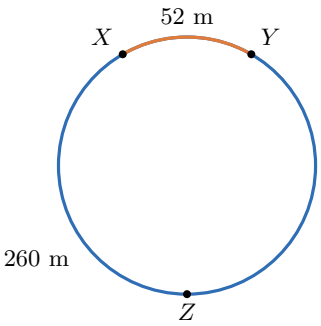
(C) $\frac{10}{3}$

(D) $\frac{15}{2}$

PART B · SELF-ASSESSMENT — 21 PROBLEMS

36. MEDIUM · DESMOS

Points X, Y, and Z lie on a circular fountain, as shown below. On this circle, minor arc XY has a length of 52 meters, and major arc XZY has a length of 260 meters. What is the circumference, in meters, of the circle?



(A) 208

(B) 260

(C) 312

(D) 52

37. MEDIUM · DESMOS

What is the center of the circle in the xy-plane defined by the equation $x^2 + y^2 - 14x + 22y + 10 = 0$?

(A) $(-7, -11)$

(B) $(7, -11)$

(C) $(7, 11)$

(D) $(-7, 11)$

38. Medium · Desmos

A circle in the xy-plane has equation $(x+2)^2+(y-5)^2=49$. What is the radius of the circle?

(A) 14

(B) 24.5

(C) 49

(D) 7

39. Medium · Desmos

A circle has a radius of 9 cm. What is the length, in cm, of an arc that subtends a central angle of 80°? (Use $\pi \approx 3.14$.)

(A) 12.6

(B) 15.7

(C) 18.8

(D) 9.4

40. Medium · Desmos

The equation of a circle in the xy-plane is $x\,(x-8)+y\,(y+6)=39$. What is the radius of the circle?

(A) 7

(B) 8

(C) 9

(D) 6

41. Medium · Desmos

A circle has an area of 144π ft^2. What is the circumference of the circle?

(A) 12 ft

(B) 12π ft

(C) 24π ft

(D) 48π ft

42. Medium · Desmos

A circle has radius $2x - 3$ and circumference $(3x + 4)\pi$. What is the area of the circle?

(A) 289π

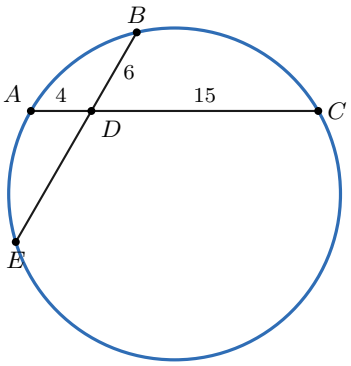
(B) 17π

(C) 34π

(D) 196π

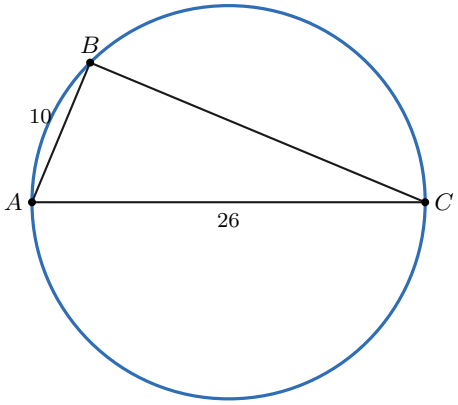
43. Medium · Desmos

Chords $\overline{AC}$ and $\overline{BE}$ of a circle intersect at point D inside the circle. If $AD = 4$, $DC = 15$, and $BD = 6$, what is the length of $\overline{BE}$?



44. Medium · Desmos

Segment AC is a diameter of a circle, and point B lies on the circle. If $AC = 26$ and $AB = 10$, what is the length of BC?



(A) $\sqrt{776}$

(B) 26

(C) 16

(D) 24

45. Medium · Desmos

The equation $x^2 - 18x + y^2 - 6y = -38$ represents a circle in the xy-plane. What is the area of the circle?

(A) $2\sqrt{13}\,\pi$

(B) 43π

(C) 52π

(D) 128π

46. Medium · Desmos

A circle has radius 15. An arc of the circle has length 6π. What is the measure, in degrees, of the central angle that subtends the arc?

(A) 18

(B) 72

(C) 36

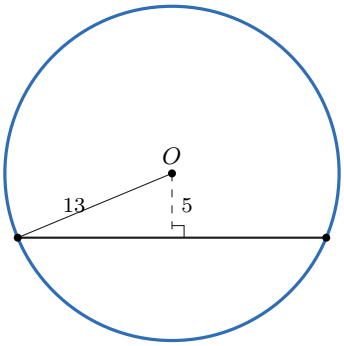
(D) 144

47. Medium · Desmos

A sector of a circle with radius 6 has area 12π. What is the measure, in degrees, of the central angle of the sector?

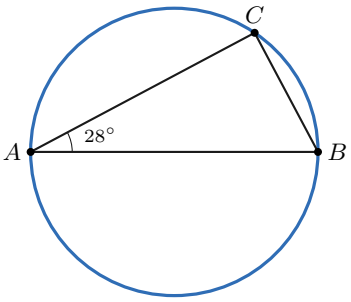
48. Medium · Desmos

A chord of a circle with radius 13 is 5 units from the center of the circle. What is the length of the chord?



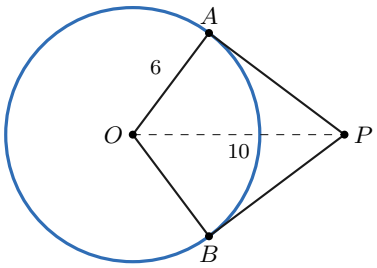
49. MEDIUM · DESMOS

Points A and B are the endpoints of a diameter of a circle, and point C is on the circle. The measure of angle CAB is 28°. What is the measure, in degrees, of angle ABC?



50. HARD · DESMOS

Point P is 10 units from the center O of a circle with radius 6. Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the circle at A and B. What is the area of quadrilateral $PAOB$?



51. Medium · Desmos

In the xy-plane, a circle has center $(-3, 5)$ and is tangent to the x-axis. Which of the following is an equation of the circle?

(A) $(x+3)^2 + (y-5)^2 = 9$

(B) $(x+3)^2 + (y-5)^2 = 5$

(C) $(x-3)^2 + (y+5)^2 = 25$

(D) $(x+3)^2 + (y-5)^2 = 25$

52. Medium · Desmos

In the xy-plane, the line $y = 6$ intersects the circle $x^2 + y^2 = 100$ at two points. What is the distance between the two points?

53. Medium · Desmos

An arc of a circle has length 10π centimeters and subtends a central angle of $\dfrac{2\pi}{3}$ radians. What is the radius of the circle, in centimeters?

54. Medium · Desmos

Two concentric circles have the same center. The outer circle has radius 7, and the area of the region between the two circles is 48π. What is the radius of the inner circle?

55. Medium · Desmos

A sector of a circle has radius 12 and perimeter $24 + 4\pi$. What is the measure, in degrees, of the central angle of the sector?

(A) 60

(B) 120

(C) 240

(D) 30

56. Hard · Desmos

A circle has area 50π. A square is inscribed in the circle so that all four vertices lie on the circle. What is the area of the square?